

THE ROLE OF ELECTRON-ELECTRON INTERACTIONS IN CONDUCTING POLYMERS

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ABSTRACT

A few general concepts of the many-electron problem and some specific effects of Coulomb interactions in conjugated polymers are summarized. The emphasis is on key experiments, mostly on the best studied material polyacetylene.

INTRODUCTION

Electrons are charged particles and interact among themselves via the Coulomb repulsion. The effects are especially important in partly filled narrow bands as for instance in the transition metals or in heavy fermion compounds. During the years it has become more and more clear that electron-electron interactions play also essential roles in π -conjugated polymers, although the measurable effects are more subtle than, say, in band ferromagnetism. The purpose of this contribution is to present a few basic concepts and theoretical methods and to discuss some clear signatures for interaction effects, in particular for the archetypal material polyacetylene. For more details the reader is referred to a recent review paper [1].

MICROSCOPIC HAMILTONIAN

Consider a collection of nuclei at positions $\vec{R}$ and with charge Ze and of N electrons at positions $\vec{r}_i$ and with momenta $\vec{p}_i$, $i = 1, \dots, N$. The fundamental microscopic Hamiltonian is given by

$$H = \sum_{i=1}^N \left(\frac{p_i^2}{2m} - \sum_{\vec{R}} \frac{Ze^2}{|\vec{r}_i - \vec{R}|} \right) + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\vec{r}_i - \vec{r}_j|}, \quad (1)$$

where we have omitted the kinetic energy of the ions. Even for simple materials it is hopeless to try to determine exactly the many-electron eigenstates of the Hamiltonian (1). Usually one of two essentially different, but somehow complementary approximation schemes are chosen. The first, commonly referred to as *ab initio* method, attacks the full microscopic problem, but uses (drastic) approximations for calculating energies, charge densities, transition matrix elements and so on. The second scheme first tries to simplify the problem, for instance by reducing

the Hilbert space to a hopefully more tractable subspace, and then studies the resulting model Hamiltonian.

ELECTRON CORRELATION

A widely used concept, although “something like a misnomer”[2], is that of electron correlation. In order to clarify its meaning we look at the simple jellium model where the ions at positions $\vec{R}$ are replaced by a homogeneous medium with constant positive charge density. This model is less important for its application to real materials (except maybe for simple metals) than for its role as a generic many-body problem. It contains only one relevant parameter, the quantity r_s/a_0 where a_0 is the Bohr radius and the length r_s is related to the electron density n by $(4\pi/3)r_s^3n = 1$. A systematic expansion of the ground state energy in powers of r_s/a_0 yields

$$\frac{E}{N} = \left\{ \frac{2.21}{(r_s/a_0)^2} - \frac{0.916}{(r_s/a_0)} + 0.0622 \ln \left(\frac{r_s}{a_0} \right) - 0.096 + \dots \right\} Ry, \quad (2)$$

where $Ry = e^2/(2a_0)$ is the Rydberg constant [3]. The first two terms are, respectively, the kinetic energy and the exchange term of the Hartree-Fock approximation (the Hartree term is exactly cancelled by the interaction with the positive background). The remaining terms, i.e., everything beyond Hartree-Fock, is referred to as the correlation energy.

The questionable choice of this name can be illustrated by the pair distribution function $g_{\sigma\sigma'}(r)$, which is proportional to the probability of finding two electrons with spins σ and σ' at a distance r . In Hartree-Fock approximation $g_{11}(r)$ vanishes for $r \rightarrow 0$ (a consequence of the Pauli principle, one speaks of an “exchange hole”), while $g_{11}(r)$ is independent of r (there is no “correlation hole”). Therefore (statistical) correlations are already taken into account by the Hartree-Fock approximation.

LANDAU'S THEORY OF THE FERMI LIQUID

The expansion parameter r_s/a_0 is of the order of 4 even in the case of alkali metals and therefore the expansion (2) is of dubious validity. Nonetheless experiments show that many properties, for instance the magnetic susceptibility or the specific heat, are well described by a non-interacting (or weakly interacting) gas of fermions. This puzzle has led L.D. Landau to propose his famous theory of the Fermi liquid [4]. The basic idea is not to worry so much about the ground state (which is anyhow difficult to characterize experimentally), but rather to consider the low-lying excitations. Landau's fundamental postulate is the one-to-one correspondence between the low-lying excitations of the (strongly) interacting electron system and the low-lying excitations of the non-interacting system. Thus he replaces the strongly interacting particles by weakly interacting quasi-particles, described by an effective mass m^* and additional parameters (“Landau parameters”) for the scattering processes between these quasi-particles. This theory has not only been very successful in describing the low temperature properties of simple metals, but it can also be used for systems where its applicability is not a priori obvious such as the heavy fermion compounds [5]. It was Landau himself who justified his theory by a detailed analysis of perturbation theory [6].

It is important to reemphasize that Landau's approach is valid only for low-energy excitations and therefore cannot be used for describing optical absorption or photoemission experiments. Moreover, in its original form the theory is only applicable to "normal Fermi liquids". Nevertheless it is of fundamental relevance also for superconductors, as it allows to separate the (perturbative) effects of the strong Coulomb repulsion from the (non-perturbative) effects of the phonon-induced electron-electron attraction. It is in fact the central assumption of BCS theory that in a first step one can get rid of the strong repulsive interactions by replacing particles by quasi-particles with an effective mass m^* and in a second step treat the residual attractive interaction explicitly [7]. This procedure is empirically justified in BCS superconductors where the effects of the residual interactions are limited to the region close to the Fermi energy, opening an energy gap of the order of 1 meV. We notice that this separation is much harder to justify in high-temperature superconductors, where the coherence length (essentially the ratio between bandwidth and bandgap) is rather small. In fact it has even been questioned whether the normal phase of these materials can be described as a normal Fermi liquid [8].

CONJUGATED POLYMERS

In conjugated polymers the low-energy electronic excitations are associated with π electrons. There is no doubt: the bare interactions between π electrons are not weak. To characterize the relevant energies one usually starts from a representation in terms of atomic orbitals (or related Wannier functions), which is complementary to the nearly free electron picture used for describing simple metals. In the Wannier representation the relevant parameter is the ratio between the on-site Coulomb interaction U and the resonance (or hopping) integral t , and it is generally accepted that $U/t \approx 4$ in conjugated polymers. It is also well-known that a one-dimensional metal is unstable with respect to a lattice distortion and a concomitant opening of a gap at the Fermi energy. This "Peierls instability" is a typical one-dimensional phenomenon, but otherwise has much in common with superconductivity. Therefore it is quite natural to try to use a two-step procedure for describing this instability, very much like that of BCS theory described above. Such a procedure was in fact proposed by Heeger and coworkers [9]. They suggest to renormalize first the strong bare interactions by mapping the microscopic π -electron Hamiltonian (implicitly) on that of a system of quasi-particles with weak residual interactions. Subsequently one should treat the non-perturbative Peierls instability (due to the electron-lattice couplings) explicitly, including the residual electron-electron interactions as a perturbation.

The idea of a Fermi liquid theory for conjugated polymers, as attractive as it may appear at a first glance, cannot be maintained for the following reasons. First, the observed gap (about 1.8 eV in polyacetylene) is large. Second (as is also acknowledged in Ref. 9), electron-electron interactions are non-perturbative in one dimension. Third, electron-lattice (electron-phonon) and electron-electron interactions act cooperatively in the Peierls instability. Therefore the situation is not dissimilar to that encountered in the problem of high-temperature superconductivity; a quantitative description appears to require an explicit treatment of the strong

electron-electron interactions. This does not mean that effective single-particle models are useless. Indeed for many purposes it will still be sufficient to include the interaction effects within a renormalized parameter. But one should not be surprised if different experiments require different parametrizations. There are however several experimental results where a simple rescaling of a parameter does not work, as will be discussed in what follows.

EXPERIMENTAL SIGNATURES

Signatures for electron-electron interaction effects in conjugated polymers may be classified into two groups, strong signatures where experimental results disagree qualitatively with the predictions of effective single-particle theories, and weak signatures where measurements are quantitatively at odds with numbers predicted by theories neglecting explicit electron-electron interactions. A brief summary is given here, however without attempting completeness nor presenting any details.

Among the strong signatures the best studied is undoubtedly the sequence of excited states in polyenes. In effective single-electron theories (as well as in Hartree-Fock approximation for interacting π electron models) the lowest excited state is dipole-allowed (the 1^1B_u state), whereas there is ample experimental evidence that in polyenes the lowest excited state is dipole-forbidden (the 2^1Ag state) [10]. This inverted ordering of excited states is very natural within models involving (not too weak) electron-electron interactions [1,11]. The extrapolation of finite chain calculations to an infinitely long chain preserves this ordering [12]. For the (infinitely long) Hubbard chain this is well understood as there is a continuum of (dipole-forbidden) magnetic excitations starting at zero energy, whereas the charge excitations have a gap [1,13].

A second strong signature is provided by magnetic resonance studies, in particular ENDOR and triple resonance experiments [14,15]. In these experiments the spin densities, attributed to π electron bond alternation defects [16], can be determined by analyzing the hyperfine splittings. In Hückel theory (or within the equivalent SSH model [17]) the spin densities of such extended defects ("neutral solitons") are positive on, say, even sites and vanish on odd sites, whereas in models including electron-electron interactions the spin densities are positive on even and negative on odd sites (or vice versa). This represents an interesting consequence of an incipient instability towards a broken symmetry ground state of the form of a spin-density wave (SDW). Unlike some organic conductors [18] conjugated polymers prefer another broken symmetry ground state, the bond-order wave (BOW) or bond alternation, accompanied by lattice dimerization. Therefore, in the homogeneous ground state the SDW is suppressed. However, in inhomogeneous states, where the BOW is locally suppressed, a local SDW can appear, leading to an alternating spin density over the extent of the defect. The coexistence of positive and negative spin densities has been clearly observed in magnetic resonance experiments [14,15], and the interpretation of the data in terms of alternating spin densities over an extended defect yields a ratio of 0.43 between negative and positive contributions [15]. Such a ratio agrees with calculations for intermediate interaction strength ($U \approx 7\text{eV}$ and $t \approx 2.5\text{eV}$ [19]).

A third set of data giving strong evidence for interaction effects is provided by photogeneration experiments [20]. Within Hückel theory electron-hole symmetry implies that a soliton

defect induces an electronic level exactly at the middle of the (Peierls) gap, independent of its charge. The expected creation of pairs of solitons after photoexcitation [9] therefore should yield additional optical absorption at half the optical gap. Instead, it is found that transitions involving charged solitons are shifted downwards to about 0.45 eV, whereas a neutral state induces a high-energy absorption around 1.35 eV [21,22]. The high-energy peak has been a puzzle for a long time. Recently magneto-optical studies have shown quite convincingly that the neutral state responsible for this transition is a neutral soliton [23]. The down-shift of transitions involving charged solitons and the up-shift of transitions induced by neutral solitons can be easily understood as an effect of electron-electron interactions. The large splitting of about 0.9 eV points to a rather large value of U [24], consistent with the value deduced from ENDOR data.

An argument showing that Hückel theory fails quantitatively in the case of conjugated polymers has been advanced by Ovchinnikov and coworkers [25]. If one determines the Hückel parameters on the basis of vibrational and structural properties of small organic molecules, one realizes that the dimensionless parameter λ entering the expression for the energy gap

$$E_g = 16t \exp \left\{ - \left(1 + \frac{1}{2\lambda} \right) \right\} \quad (3)$$

is a factor of 2 too small to account for the extrapolated gap of an infinitely long chain (or the observed gap in polyacetylene) [26]. Therefore these authors proposed that the conjugated polymers should be viewed as Mott-Hubbard insulators instead of Peierls semiconductors. In fact, a Hubbard chain (for an average density of one electron per site) exhibits an optical gap

$$E_g = \frac{8}{\pi} \sqrt{U} t \exp \left\{ - \frac{2\pi t}{U} \right\} \quad (4)$$

(for not too large U) [13], and with the values for U and t quoted above the measured gap of about 1.8 eV can be reasonably well reproduced. One may object that a simple way to save the Peierls mechanism would be to use an effective electron-phonon parameter λ in Eq. (3), in particular since taking over parameter values from small molecules is not free of uncertainties. However, one immediately runs into trouble with respect to vibrational properties. In fact, for a dimerized chain described by Hückel theory the ratio between the effective carbon-carbon force constant K_{eff} (including the screening by π electrons) and the bare (σ bond) force constant K is given by [1,9]

$$K_{eff}/K = 2\lambda \quad (5)$$

This ratio is about 0.75 for polyacetylene [27], thus requiring an electron-phonon parameter $\lambda \approx 0.4$. On the other hand, in order to reproduce the optical gap on the basis of Eq. (3) one needs a value $\lambda \approx 0.2$. A consistent picture can be obtained by taking into account both the electron-phonon coupling (with $\lambda \approx 0.1$) and an on-site Coulomb term (with $U/(4t) \approx 0.7$) [28].

Above, we have discussed the shifts produced by electron-electron interactions in the case of soliton-induced transitions. Solitons occur in polymers with a doubly degenerate groundstate

such as *trans*-polyacetylene, not however in polymers with a non-degenerate ground state such as polythiophene or poly (phenylenevinylene). Nevertheless photo-induced absorption is also observed in these materials; it consists of two peaks of about equal intensity in the gap [20]. The two lines are commonly attributed to bipolarons, which may be viewed as confined pairs of equally charged solitons. Similarly as in the case of polyacetylene the lines are shifted with respect to the predictions of Hückel theory, and this shift can again be explained as a signature of electron-electron interactions [20,29]. Another discrepancy between experiment and single-particle theory is however much more pronounced. Within the SSH model the low-energy peak should have much larger intensity than the high-energy peak [30], in contrast to experiment. This discrepancy can again be attributed to the neglect of (explicit) electron-electron interactions. Indeed, similarly to the local SDW induced at a neutral soliton (as discussed above), a local charge-density wave (CDW) is expected to be generated by Coulomb interactions at a charged soliton [31] (or at a bipolaron). Such a CDW breaks locally the electron-hole symmetry and thus can drastically change the intensity ratio and remove the discrepancy between theory and experiment. We notice, however, that other mechanisms can also break the electron-hole symmetry, and therefore the intensity anomaly *per se* is not a strong signature for Coulomb effects [1].

A rather subtle interaction effect has been investigated by Fink and Leising [32]. Consider a collection of (uncoupled) polymer chains, each of them being described by a π -electron Hamiltonian of the form

$$H = H_0 + H_{int} \quad (6)$$

where H_0 describes the hopping between neighboring sites along the chains and H_{int} stands for Coulomb interactions of arbitrary range. The sum rule [33,34]

$$\int_0^\infty \sigma(\omega) = -\frac{\pi e^2 a}{2d^2 \hbar^2} \frac{\langle H_0 \rangle}{N} \quad (7)$$

relates the (real part of) the frequency-dependent conductivity $\sigma(\omega)$ to the expectation value of the hopping term $\langle H_0 \rangle$. The constants in Eq. (7) are the electron charge e , the lattice constant along the chains a , the interchain distance d and the number of sites per chain N . For the simple Hubbard model (without lattice dimerization) – $\langle H_0 \rangle$ can be calculated exactly [35]; it is a monotonously decreasing function of U and tends to zero for $U \rightarrow \infty$. The experiments [32] indeed indicate that the intensity of $\pi - \pi^*$ transitions is reduced by about 30% from what would be obtained on the basis of Hückel theory. The reduction factor is consistent with a ratio $U/(4t) \approx 0.7$. It shows that a rather large value for U is needed to account for a relatively small effect.

AB INITIO METHODS AND MODEL HAMILTONIANS

Above we have argued mostly in terms of a few essential parameters such as the resonance integral t , the on-site Coulomb term U and the electron-phonon parameter λ . One may object that these arguments are strongly linked to an over-simplified model and a real theory should start from scratch, i.e. try to deal with the microscopic Hamiltonian (1) including all

electronic and nuclear coordinates. During the last decade many *ab initio* calculations have in fact been performed, some for an individual polymer chain, some for crystalline polymers. Unfortunately, it is impossible to solve this complicated many-body problem to a well-defined degree of accuracy, and one is forced to use rather drastic approximations. The two most popular schemes have been the Hartree-Fock (HF) approximation [36] and the local density approximation (LDA) of the density functional formalism [37].

By construction the HF scheme yields the best effective single-particle approximation for the full problem. It includes both the average Coulomb potential acting on an electron due to all the other particles (Hartree term) and the (non-local) exchange potential (Fock term). The HF approximation, though computationally already quite involved, can be used as a starting point for further systematic improvements (configuration interaction). The LDA is computationally less intensive, as it includes both exchange and correlation effects approximately within a local potential. However, while the HF band structure has a well-defined meaning, the LDA bands are associated with a somewhat mysterious auxiliary fermion system.

Both schemes appear to be quite reliable for calculating ground state energies and overall geometries, but both run into problems when dealing with excited states. In HF approximation the electronic gap for polyacetylene turns out to be of the order of 5eV, far above the experimental value of 1.8 eV. This is clearly due to the neglect of correlation between electrons with different spins [36,38]. In LDA the gap is usually appreciably smaller than the experimental value, an effect which is also well understood (and partly curable [37]).

Both the large computational effort involved in *ab initio* calculations and the desire of identifying the basic variables and interactions motivate the study of model Hamiltonians. An example is the π -electron Hamiltonian (6) with an interaction term

$$H_{\text{int}} = UD + V \sum_i n_i n_{i+1} \quad (8)$$

where n_i measures the number of π electrons at site i and D the number of doubly occupied sites. Instead of considering the interaction strengths U and V as adjustable parameters, they can also be determined *ab initio*, as has been demonstrated recently, giving $U = 11.5 \text{ eV}$, $V = 2.4 \text{ eV}$ [39]. The idea of incorporating the results of *ab initio* calculations within a simplified model is also the basic ingredient of the “valence effective Hamiltonians” [40].

DIMERIZATION

The problem whether a long polyene chain spontaneously dimerizes and forms an alternating sequence of long and short bonds has occupied both theorists and experimentalists since the late thirties. Recently, once bond alternation was experimentally established in polyacetylene, the question was rephrased from “Does it ?” to “Why does it dimerize ?”. In the Hückel (or SSH) model it is clearly the Peierls mechanism which is at the origin of bond alternation. In contrast to some statements that Coulomb interactions would counteract dimerization, it is now widely accepted that moderate electron-electron interactions enhance bond alternation. Some authors even believe that these interactions dominate in provoking an electronic BOW

instability, while the lattice dimerization due to the electron-phonon coupling simply further stabilizes this broken symmetry state [41].

In order to get an idea of the main effects we look at the interaction term (8). In (restricted) HF approximation the interaction energy consists of a Hartree term depending only on the expectation value of the density, $\langle n_i \rangle$, and the exchange (or Fock) term

$$E_x = -\frac{1}{2}V \sum_i p_{i,i+1}^2 \quad (9)$$

where $p_{i,i+1}$ is the bond order. This negative exchange energy can be lowered by admitting an alternating bond order, i.e.,

$$p_{i,i+1} = p_0 + (-1)^i p_1 \quad (10)$$

Therefore the non-local exchange term favors bond alternation. The U term, generally much larger than the nearest neighbor Coulomb interaction, does not contribute to bond alternation in HF approximation. Therefore one has to go beyond HF and calculate the correlation energy. It is generally found that for not too large values of U the correlation energy favors bond alternation [1]. This can be understood as follows. Any many-electron state can be written as a superposition of real-space configurations or, more conveniently, in terms of valence bond diagrams [42]. These diagrams consist of lines between lattice sites, representing singlet pairs (covalent bonds), and of dots and crosses representing empty and doubly occupied sites. It is quite obvious that empty and doubly occupied sites are very efficient in interrupting a given sequence of nearest neighbor bonds (they act like domain walls in an Ising ferromagnet). With increasing U configurations with a large number of doubly occupied sites loose weight in the ground state, as their energy increases. This effect amounts to the deepening of the correlation hole. It is taken into account in a very transparent way by the Gutzwiller Ansatz [43,44]

$$|\psi\rangle = e^{-\eta D} |\psi_0\rangle \quad (11)$$

where $|\psi_0\rangle$ corresponds to a single Slater determinant and η is a variational parameter. It is found that η increases monotonously with U and thus suppresses more and more configurations with double occupancies.

The dimerization has also been studied extensively with *ab initio* methods. The HF approximation yields surprisingly good agreement with experiment [45], correlation effects reducing slightly the dimerization [38]. On the other hand, several LDA calculations resulted in much smaller dimerization [46,47,48]. This might be attributed to the complete neglect of non-local exchange; however, the situation is confusing as other LDA calculations agree quite well with experiment [49,50,51].

EXCITONS AND NONLINEAR OPTICS

The exciton, an electron-hole pair bound by Coulomb forces, has been extensively studied in the case of conventional semiconductors. For a uniaxial semiconductor with reduced masses $\mu_\perp, \mu_\parallel$ and dielectric constants $\epsilon_\perp, \epsilon_\parallel$ for directions perpendicular and parallel to the z axis the

important anisotropy parameter is $\gamma = \epsilon_{\perp}\mu_{\perp}/(\epsilon_{\parallel}\mu_{\parallel})$ [52]. For the particular case $\gamma = 1$ the exciton binding energy is given by $(\mu_{\perp}/m_0\epsilon_{\perp}\epsilon_{\parallel}) Ry$, where m_0 is the free electron mass. The exciton size in z direction is by a factor $(\mu_{\perp}/\mu_{\parallel})^{1/2}$ larger than in x and y direction. It is not clear to what extent this simple effective mass approximation is applicable to conjugated polymers [53], but it may serve as a very rough order of magnitude estimate. One expects $\epsilon_{\parallel}$ to be much larger than $\epsilon_{\perp}$ and $\mu_{\parallel}$ to be much smaller than $\mu_{\perp}$. Therefore the assumption $\gamma \approx 1$ is not unreasonable. The binding energy is then large for systems with a large $\mu_{\perp}$ and small dielectric constants. This may explain why excitons have been clearly observed in polydiacetylene, while their detection in polyacetylene has been elusive so far.

A recent calculation for an effective π -electron Hamiltonian (with long-range Coulomb interactions of the PPP form) has revealed that the lowest (singlet) exciton state carries most of the intensity of the linear optical absorption [54]. It will therefore also play a significant role as an intermediate state in the third-order nonlinear optical response of conjugated polymers, a research topic of high current interest. However, besides the exciton a two-photon allowed highly correlated state (of 1A_g symmetry) appears also to be strongly involved in the third order processes [55,56]. All these calculations, including explicit electron-electron interactions, indicate that the agreement between experiment and Hückel theory [57] may be fortuitous [54].

HIGHLY CONDUCTING REGIME

Up to this point the discussion was mostly concerned with the case of a half-filled π electron band representing, e.g., pristine polyacetylene with a large optical gap and a small electrical conductivity. The question to what extent electron-electron interactions are involved in the highly conducting regime is presently not understood. But it is quite clear that the effects will be fundamentally different away from half filling as compared to the case of a half-filled band. This can be seen by looking at the momentum distribution function of the Hubbard chain in these two cases [58,59]. At half filling the one-dimensional Hubbard model represents an insulator, away from half filling a "Luttinger liquid" [8].

If the disorder, e.g. due to dopant ions, is large enough to localize the wavefunctions close to the Fermi energy, the Coulomb repulsion can strongly affect the magnetic susceptibility. This can be demonstrated on the basis of a simple model [60], where the cooperative effects of electron-electron interactions and disorder yield a superposition of a Curie and a Pauli susceptibility. This simple model apparently describes rather well certain experiments [61].

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